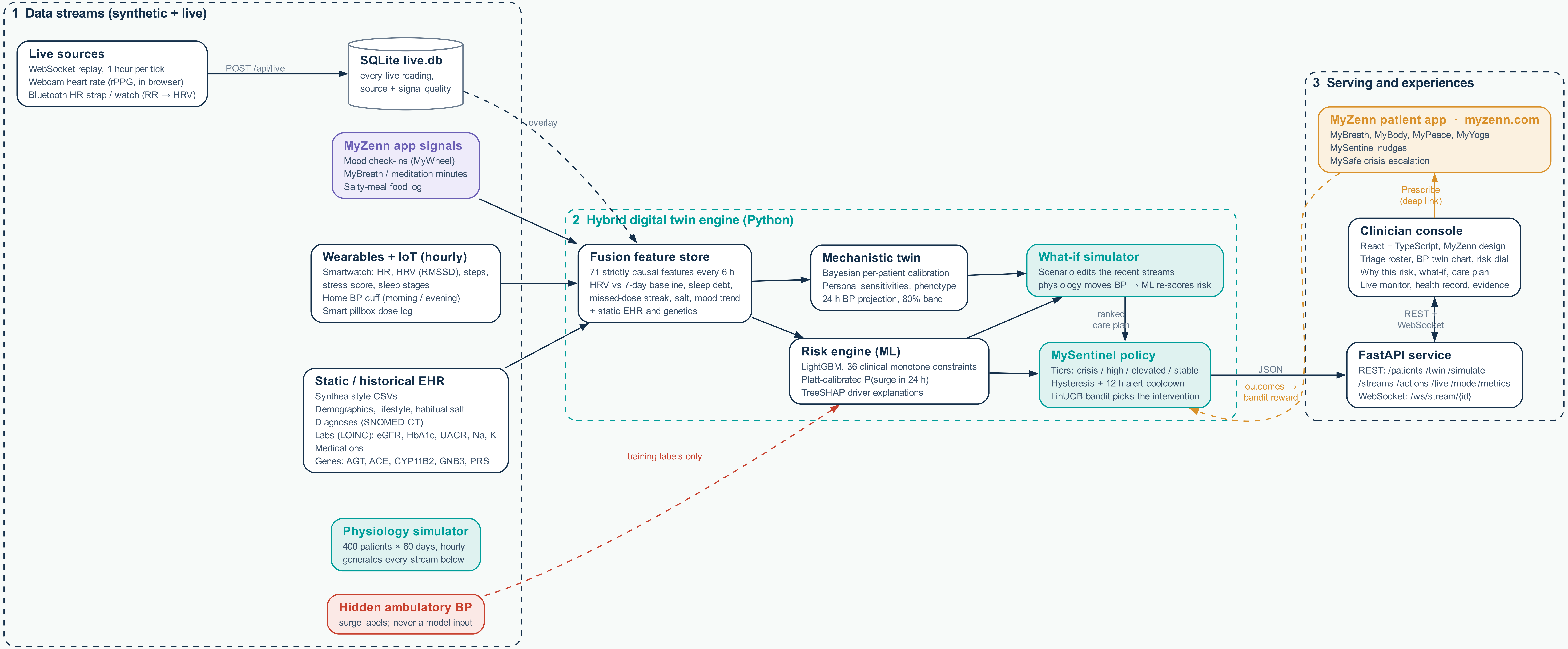
HridayTwin: system architecture

Two data streams fused into a hybrid digital twin; a doctor-facing console; interventions delivered through MyZenn.



1 Data streams (synthetic + live)

Live sources
WebSocket replay, 1 hour per tick
Webcam heart rate (rPPG, in browser)
Bluetooth HR strap / watch (RR → HRV)

SQLite live.db
every live reading,
source + signal quality

MyZenn app signals
Mood check-ins (MyWheel)
MyBreath / meditation minutes
Salty-meal food log

Wearables + IoT (hourly)
Smartwatch: HR, HRV (RMSSD), steps,
stress score, sleep stages
Home BP cuff (morning / evening)
Smart pillbox dose log

Static / historical EHR
Synthea-style CSVs
Demographics, lifestyle, habitual salt
Diagnoses (SNOMED-CT)
Labs (LOINC): eGFR, HbA1c, UACR, Na, K
Medications
Genes: AGT, ACE, CYP11B2, GNB3, PRS

Physiology simulator
400 patients × 60 days, hourly
generates every stream below

Hidden ambulatory BP
surge labels; never a model input

2 Hybrid digital twin engine (Python)

Fusion feature store
71 strictly causal features every 6 h
HRV vs 7-day baseline, sleep debt,
missed-dose streak, salt, mood trend
+ static EHR and genetics

Mechanistic twin
Bayesian per-patient calibration
Personal sensitivities, phenotype
24 h BP projection, 80% band

What-if simulator
Scenario edits the recent streams
physiology moves BP → ML re-scores risk

Risk engine (ML)
LightGBM, 36 clinical monotone constraints
Platt-calibrated P(surge in 24 h)
TreeSHAP driver explanations

MySentinel policy
Tiers: crisis / high / elevated / stable
Hysteresis + 12 h alert cooldown
LinUCB bandit picks the intervention

3 Serving and experiences

MyZenn patient app · myzenn.com
MyBreath, MyBody, MyPeace, MyYoga
MySentinel nudges
MySafe crisis escalation

Clinician console
React + TypeScript, MyZenn design
Triage roster, BP twin chart, risk dial
Why this risk, what-if, care plan
Live monitor, health record, evidence

FastAPI service
REST: /patients /twin /simulate
/streams /actions /live /model/metrics
WebSocket: /ws/stream/{id}